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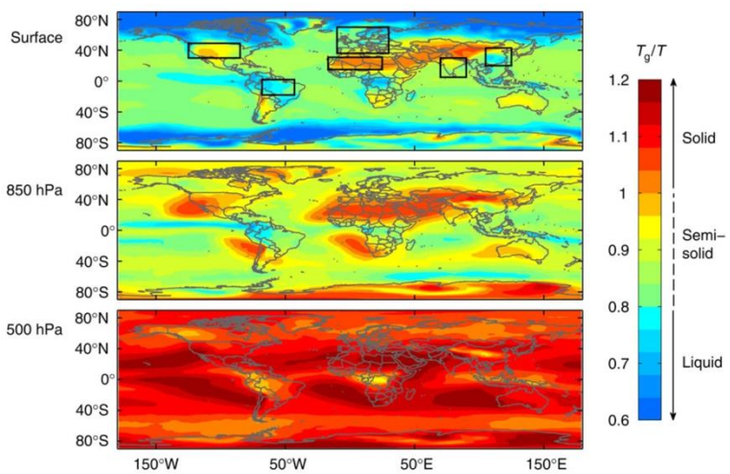
Do Glasses Cool the Earth?

From aerosol phase to anvil clouds – direct and indirect radiative impact

Where do glassy aerosols exist?

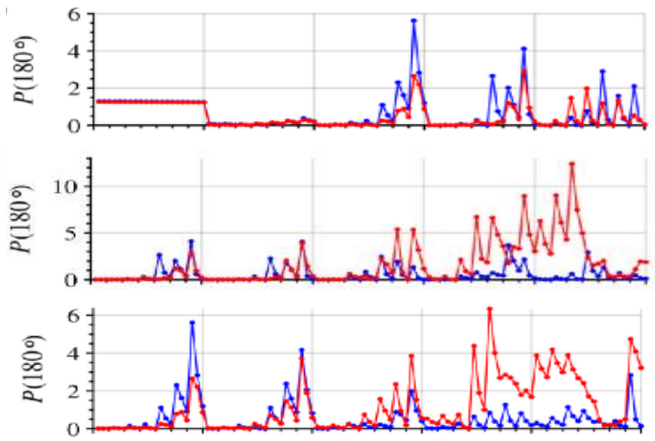
Glassy particles are expected in large parts of the atmosphere, especially at upper levels relevant for ice clouds.



Shiraiwa et al., 2017

Does particle phase affect radiation?

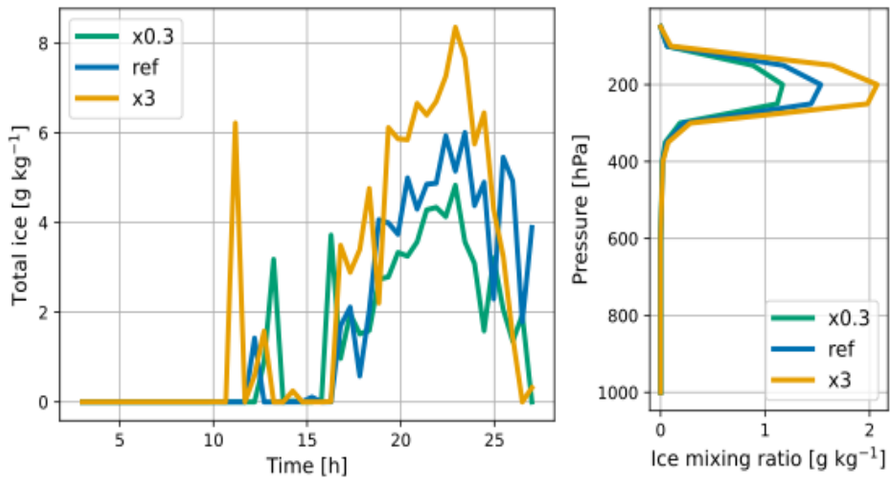
Glassy and crystalline particles scatter radiation differently, meaning aerosol phase impacts the radiative balance.



Sela et al., 2021

How do aerosols affect anvil clouds?

Changing aerosol concentration alters ice content in deep convection, but the role of aerosol type remains unclear.



So, what's missing?

We know that:

- Glassy aerosols are abundant
- Directly affect radiation
- Aerosols influence anvil clouds

However, we don't know:

- Do glassy aerosols influence ice formation?
- Do they change anvil properties?
- Are we missing key pathways in models?

I'd be happy to chat about:

- Aerosol-cloud interaction
- Ice nucleation (PIP/SIP)
- Aerosols representation in models
- Glassy / highly porous aerosol

Next: linking glassy aerosols to ice formation, anvil lifecycle and radiation

References:
Shiraiwa, M., Li, Y., Tsimpidi, A. *et al.* Global distribution of particle phase state in atmospheric secondary organic aerosols. *Nat Commun* 8, 15002 (2017). https://doi.org/10.1038/ncomms15002
Maor Sela and Carynelisa Haspel, "Direct comparison between single-scattering properties of ordered and disordered aggregates of nano-sized scattering centers," Appl. Opt. 60, 10893-10900 (2021)